



Bypass Valve Cover Installation Tool in Oil Filters Base for 3500 Engines

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Bypass Valve Cover Installation Tool in Oil Filters	
Base for 3500 Engines	0
1.0 Introduction	1
2.0 Best Practice Description	1
3.0 Implementation Steps.....	2
4.0 Benefits	2
5.0 Resources Required.....	2
6.0 Supporting Attachments / References.....	2
7.0 Related Best Practices	4
8.0 Acknowledgements	4

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1.0 Introduction

Current process for installation of the 3500 engine oil filter base valve springs (PN: 7G-6713) and the respective cover is performed manually. Compressing the spring to the closed position of the cover, requires the technician to exert a considerable amount of force (17Kg) in a small area. This task involves and requires a great amount of physical effort, and with multiple oil filter base covers/springs this task becomes uncomfortable and tiring for the technician.



Figure 1. Procedure manual installation of the cover

This Best practice aims to facilitate the development of this task to design a user-friendly process that is fast and safe to use.

2.0 Best Practice Description

This tool applies to the bases of 3500 engine oil filters. The press consists of a C-shaped piece made of 7/16" steel grade 514. Compression is performed by an endless 7/8" screw with a pitch of 7 mm, with a C-shaped tip that rotates freely to accommodate the spring covers on the oil filter bases. Also, two screws are used that are similar to the original cover screws but longer (4"), guide the cover while being compressed and prevent rotation.



Figure 2. Tool Architecture.

The tool weight is 1.7kg. The spring requires a force equal to or greater than 17 kg. For this reason this system is in the form of a press which facilitates the installation.

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3.0 Implementation Steps

- Check the status of the tool to verify the integrity before use.
- Place spring and cover fitting with guide screws. (1)
- Install the tool "C" ensuring proper engagement with the flat bottom of the base, turn the screw (down) until the tool is self-supporting the product of the force exerted by the spring (2).
- Next bring down the screw cap to its final installation position (3), then manually remove the interim guides and install the screws (bolts).
- You should remove one guide and install one bolt, then repeat for the other, so that you never have the tool on the part without a bolt or guide screw in place.
- Finally tighten to the required torque (4).



Figure 3. Procedure for operation of the tool.

- After installing the cap, release the lead screw to remove the tool

4.0 Benefits

- Easy fabrication
- Easy operation
- Improved component assembly
- Spring secure installation, reduces accidents
- Decreases fatigue avoid manual installation technician

5.0 Resources Required

Plans of the tool (see attached map)

6.0 Supporting Attachments / References

EDITOR'S NOTE: Due to the fact P/N 7G-6713 Spring can work with multiple oil bases, please make sure this tool is applicable to your application. Dimensions will vary, depending on the operation you are undergoing.

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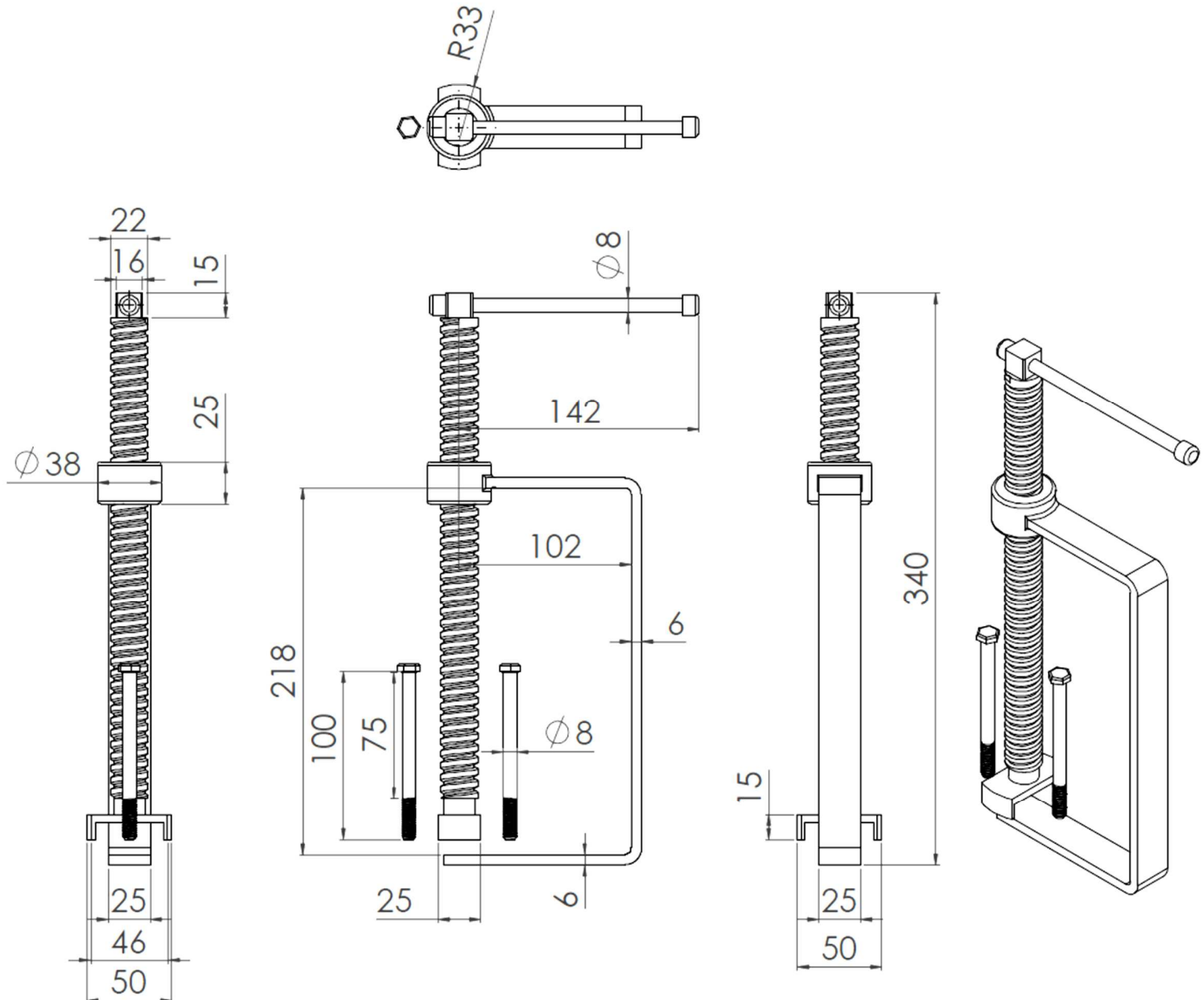
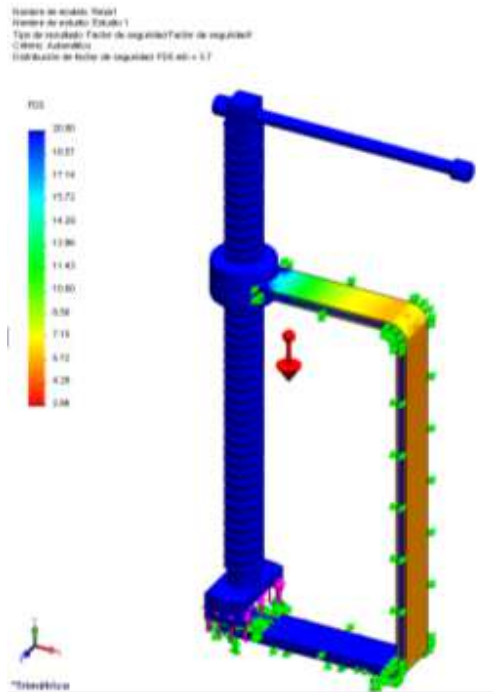


Figure 4. Map of the tool.

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		NUMBER 1501-4.03-1361	



Safety factor 3.7 with test load 18kg

7.0 Related Best Practices

None

8.0 Acknowledgements

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